Pilot on Improved fundikila integrated with *Crotalaria* as green manure for sustainable rice production in wetland

Compiled by RESCAP

Background



Chitemene system could support between 2 and 4 persons per square km depending on the amount of suitable land available. Nearly 80-90% of farmers in the high rainfall region practise Chitemene. It is said that soils in arable land depleted for sustainable crop production where the population is higher and there are fewer trees.

Fundikila is a system to culminate the breakdown of the Chitemene system as the area became deforested by an increasing population. The combination of green-manuring and alternating cereal crops with legumes under fundikila helps to slow the exhaustion of soil fertility. It has been estimated that the fundikila system is able to support 20-40 persons per square km (Mansfield et al., 1975), considerably more than the Chitemene system.

However, the organic matters incorporated into the soil are grass or weeds which mainly contain more grass materials rather than nitrogenous substance and higher content of carbon leads to nitrogen starvation resulting from decomposing process by micro organisms in the soil.

Therefore, introduction of green manuring crop in the fundikila system is ideal to increase soil quality, fertility and texture as well as structure.

What is green manure?

GREEN manuring is ploughing under or soil incorporation of any green manure crop while it is green or soon after it flowers. Green manures are forage or leguminous crops that are grown for their leafy materials needed for soil conservation. Any crop which is buried in soil with the idea to increase soil quality, fertility and texture as well as structure is termed as green manuring crop. Such crops can be grown either before or along with the main crop. These act as a reservoir of nutrients. Growing of leguminous crops increases nitrogen availability in the soil. These nutrients are released in the soil when they are ploughed.

How do we use green manure?

Burying in should be done at a time when the crop starts flowering in order to leave maximum amount of vegetative matter and contents. Green manure crops are directly ploughed into the soil about 2-3 weeks before the main crop. To get maximum benefit from green manure crops, it is important to plough them in before they begin to flower or fruit. In the case of trees, the leaves and tender branches are cut and then ploughed into the soil. Green manuring is an easy method of improving soil structure and fertility. Since the crop is grown right there where it is utilised, it makes the costs minimal.

Why Crotalaria as green manure?

Selection of green manure crops: Fast growing; producing abundant and

succulent shoots (foliages); well adapted to local condition; having a high rate of nitrogen fixation; having high water-use efficiency when used in drier regions; while still producing substantial quantities of top-growth; able to fix nitrogen in the soil. One of commonly used green manure crops is Sun hemp (Crotolaria juncea).

Many uses of Crotalaria include:



- Green manure
- Nitrogen fixation
- Weed suppression - Striga
- Livestock forage – Cattle, Chicken
- Pest control - Nematode
- Soil erosion control
- Unlike other legumes, sunnhemp performs well on poor and acidic soils
- Feed of tilapia

Rice as beneficiary crop by this method

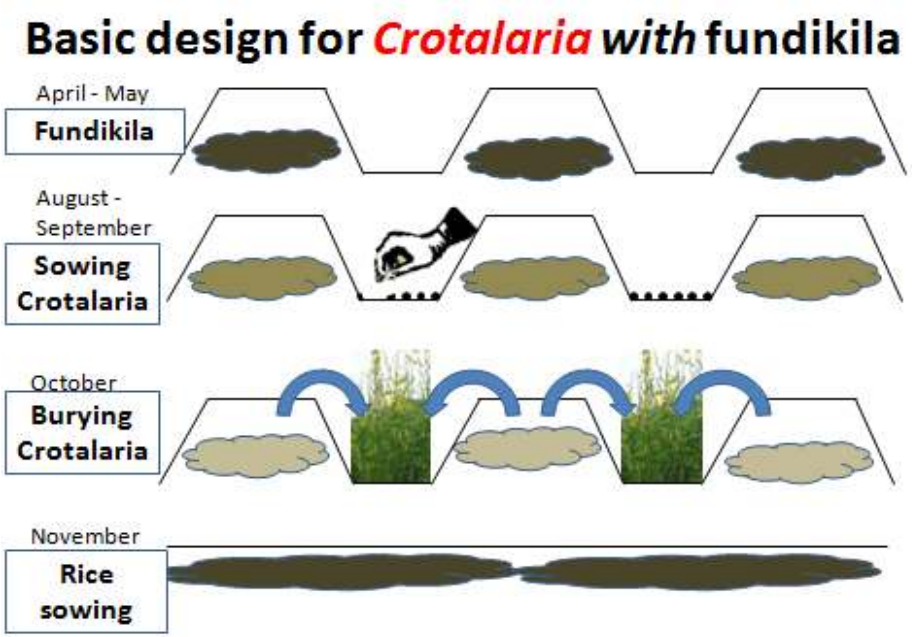
- Demand for rice is growing and it is selected as one of potential crops though Need Assessment Survey. Subject Matter Specialists in the Province suggested rice as potential crops.
- Farmers experience indicated Nerica growth under fundikila was better than without fundikila.
- In other countries, rice was grown following incorporation of green manure 45 or 55 days after sowing and rice grain yield was greater with Crotalaria.

Basic design and the procedures of improved fundikila pilot

1. Legumes such as Sunhemp are planted on field where crops will be planted
2. Legumes and grasses are buried underground as ordinal fundikila
3. Land preparation is done for rain-fed production for rice in Dambo

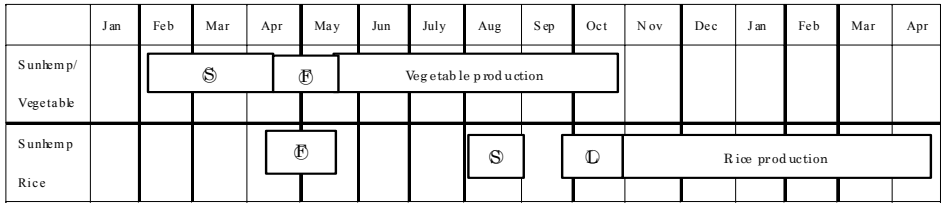
Or

1. Grasses are buried underground as ordinal fundikila
2. Legumes such as Sunhemp are planted and grown on fundikila mound
3. Sunhemp grown on the mound is incorporated into the soil during the land preparation for rain-fed production for rice in Dambo.

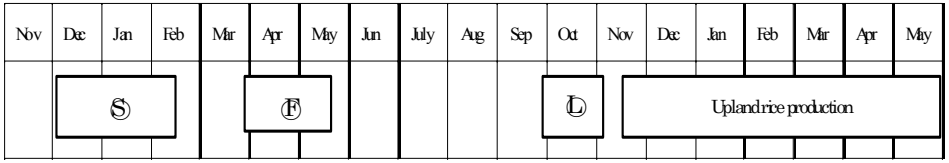


(1) Improved winter ploughing in wetland/Dambo for lowland rice/vegetable

Site selection is a critical factor to determine the success or failure of the system. The site can be a field where vegetables are grown using residual moisture from May to September or irrigation furrows are available.



(2) Improved winter ploughing in virgin upland for Maize or upland rice



Practical methods on improved fundikila on rice in wetland

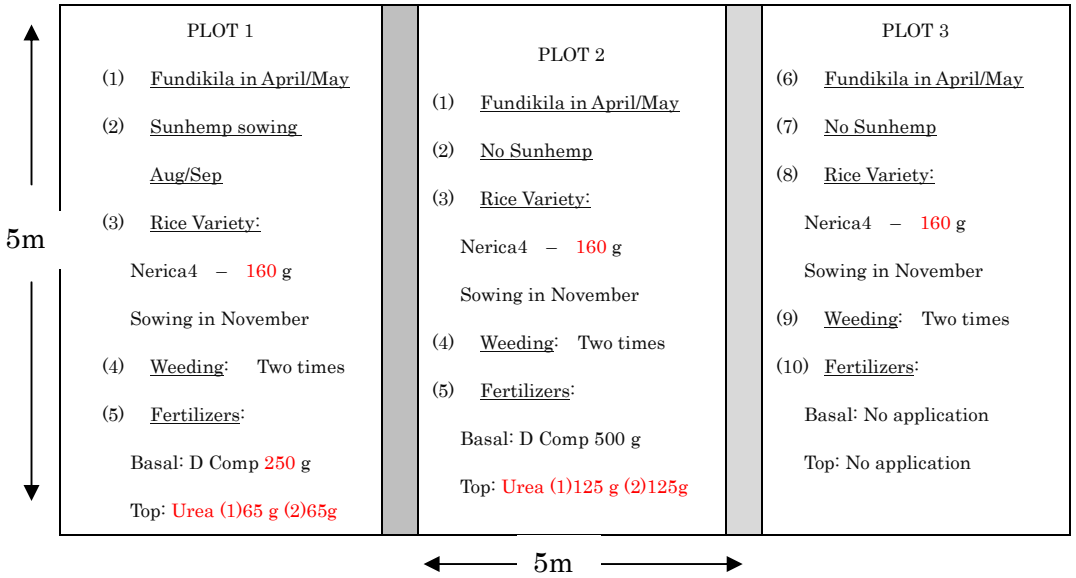
1 Site Selection, Land Demarcation and Fundikila

1.1 Select a suitable area with relatively good soils, free from natural barriers, previously cultivated and with minimum undulations;

1.2 The selected site should be in a strategic place, e.g. near a major road for easy access and passersby to observe;

1.3 Demarcate or measure the piece of land earmarked for the pilot using a field measuring tape and a rope and mark accordingly using pegs;

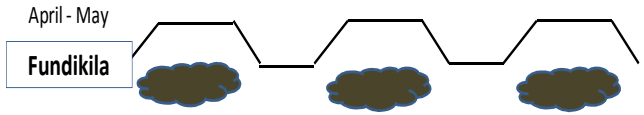
1.4 Divide the plot into three subplots as shown on the Field Layout Plan.



Fertilizer rate for the plot 1 is 50% of rates for plot 2.

Plot 3 is a common practice of small scale farmer's.

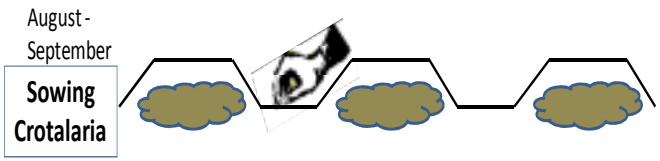
1.5 Grasses are buried underground as ordinal fundikila for both plots.



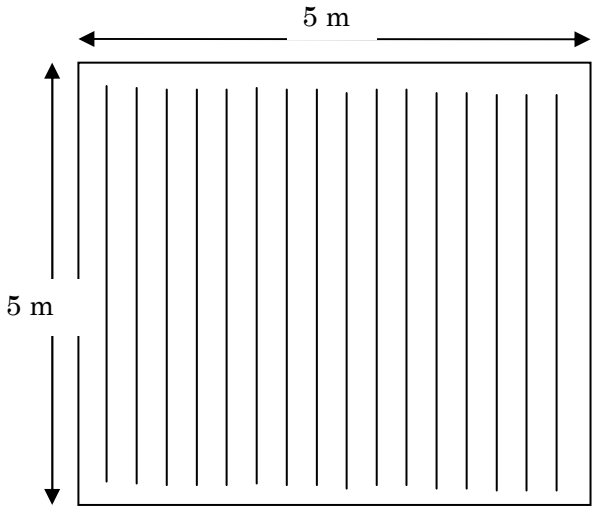
2 Planting of sunhemp

2.1 Site selection is important because sunhemp is grown in wetland using residual moisture or furrow irrigation. Avoid termite prone areas.

2.2 Sunhemp is planted in rows or broadcasted and grown in the field in August or September.



- ✓ Type of seed is Black seeded (*Crotalaria juncea*)
- ✓ Seed rate: 15kg /lima = 150g/plot
- ✓ Loosen the path ways between fundikila mounds or ridges using hoes to loosen up the soil one day before sowing.
- ✓ Then irrigate the path ways if furrows are available.
- ✓ Broad cast or drill the sunhemp seed in rows between the fundikila mounds or ridges by using line marker 30cm apart in Plots and bury accordingly.

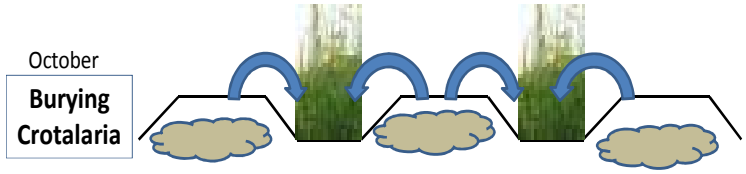


3 Land Preparation

3.1 Dig the land thoroughly using hoes to loosen up the hard pan for proper penetration of roots of the rice crop.

3.2 Biomass of sunhemp grown is incorporated into the soil while preparing land for rice planting.

3.3 Level the field smoothly using hoes so as to obtain a seed bed of fine tilth to facilitate even germination of seeds.



4 Basal-dressing Fertilizer Application and Planting

4.1 Apply D Compound fertilizer in the given quantities in Plots.

PLOT 1(with sunhemp)

Basal: D Comp 250 g/plot

PLOT 2(without sunhemp)

Basal: D Comp 500 g/plot

PLOT 3(without sunhemp without fertilizer)

4.2 Drill the rice seed sparsely (10g of seed per row = 160g of seed per plot) in the given quantities into furrows by using line marker 30 cm apart in Plots and bury accordingly (16 rows per plot).



5 Weeding and Top-dressing Fertilizer Application



5.1 Weed in Plots two weeks after emergence when the crop is at seedling stage.

5.2 Weed in Plots again six (06) or eight (08) weeks after emergence when the crop is at tillering or booting stage.

5.3 Apply Urea fertilizer in the given quantities in Plots by broadcasting evenly and incorporate into the soil.

PLOT 1(with sunhemp)

First split: 65 g/plot

Second split: 65g/plot

PLOT2(without sunhemp)

First split: 125 g/plot

Second split: 125g/plot

PLOT2 (without sunhemp without fertilizer)

6 Harvesting & Post-harvest Crop Handling

6.1 Harvest the crop plot by plot using sickles when 90% of the panicles turn golden yellow.

6.2 Dry the harvested crop plot by plot for three (03) days under shade for drying before threshing.

6.3 Thresh the crop plot by plot by beating the panicles against a hard surface using a stick, taking care not to break the grains in the process.

6.4 Winnow the grain plot by plot so as to get rid of chaff and other foreign materials such as soil, stones, etc.

6.5 Spread the grain on a mat or concrete floor plot by plot for 2 – 3 days for further drying before packing and storage.

7 Packing & Storage

7.1 After recording the quantities of the grain plot by plot, bulk and pack in appropriate packaging materials.

7.2 Store the grain in well-ventilated storage houses under lock and key.

7.3 Take care of rain water, rodents, termites and other storage pests by regularly inspecting the stores and taking appropriate measures when need arises.



DATA COLLECTION FOR FUNDIKILA WITH SUNHEMP FOR RICE FOR 2011/ 2012 SEASON

PLOT Nº.	DOP	D 50% E	D 50% F	DOH	GY	PDS
1						
2						
3						

DOP----------Date Of Planting

D 50% E----------Date of 50% Germination

D 50% F----------Days to 50% Flowering

DOH----------Date Of Harvesting

GY----------Grain Yield (Kgs)

PDS----------Pest & Disease Score:

0 – Nil;

1 – Low;

2 – Medium;

3 – High

Field note for farmers

When to do	Actual date	What to do/growth record	Check/remark
April/May		Fundikila making	
End Aug. to beginning Sep.		Sowing sunhemp seed	
Beginning November.		Bury sunhemp and land preparation for rice	
Mid. November		Seed selection and planting rice seed	
1 week after sowing		Germination	
21 days from seed emergency		Basal dressing of D-compound at specified rates then first weeding	
43 days after seed emergency		First top dressing and second weeding	
63 days after seed emergency		Second top dressing and weeding	
Mid. Apr. to May		Harvesting	